

The AUDIS Catalog of Human HRTFs

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Abstract: In the European-Union funded project AUDIS (AUditory DISplay) a multipurpose auditory display for 3-D hearing applications is being developed. A main goal is to develop a special sound generator in combination with an auditory-symbol database that can, e. g., be used in avionics and automotive industries. As the project relies heavily on binaural technology and, hence, on the availability of reliable human HRTF data, a special program for collecting these data has been undertaken. In the course of this program, a round-robin test has been performed in four different laboratories. Differences in the measured data across the different laboratories have been analysed. The analysis resulted in recommendations for common measuring procedures to be applied. A catalog of HRTFs has then been collected, and perceptual verification test have been performed. A sampler CD of the catalog will be available through the literature distribution system of the European Acoustics Association, DOCUMENTA ACUSTICA. For details please visit <<http://eaa.essex.ac.uk/eaa/>>.

DATA COLLECTION

The main body of the data has been collected at Bochum and Soesterberg, where about 20 individual sets of HRTFs each have been measured on human subjects. In the following we first describe the set-up at Bochum and, consequently, what is different at Soesterberg. To be sure, the two other labs involved have well established set-ups of their own. We only do not deal with these here, as this would not add information from a scientific point of view.

Measurements were performed in an anechoic chamber of sufficient size (about 5m x 5m x 5m) and sufficiently low limiting frequency (110Hz). The subject was seated in the middle of the room on a rotating chair. An arc of eleven loudspeakers was placed at a distance of 2.4 m from the centre of the subject's head, covering the elevation range from -10° to $+90^{\circ}$ in 10° steps. Azimuth was varied by rotating the chair in 15° steps. The position of the head was adjusted in the zero-azimuth position by means of three laser beams, pointing at the ear-canal entrances and at the tip of the nose, and then supported by a slim neck rest. The azimuth positions were verified by the subject by looking at little marks at appropriate positions. Altogether, HRTFs were measured for 122 directions of sound incidence for each of the two ears of each subject, i.e., some lateral elevations were left out.

The two microphones (Sennheiser KE4-2) were each placed at about 4mm inside the ear canals, surrounded by a ring of foam rubber (Etymotic). In this way, they were placed at a position where we already have one-dimensional sound propagation. In this way, all variance of the HRTFs caused by the direction of sound incidence is captured at the pick-up point chosen. The measuring signal was random-phase pseudo noise with a series of 4096 samples (44.1kHz sampling frequency), applied 100 times at a level of 70dB (SPL). In fact, the measuring procedure renders impulse responses from which the HRTFs are derived by Fourier transforms. Yet, prior to the transforms the impulse responses were truncated to a length of roughly 3ms. This procedure reduces the noise and eliminates possible reflections from the wall, etc. The truncating time window was properly adjusted by visual inspection in each case.

Prior to the measurements, the eleven loudspeakers were electronically equalised by determining their free-field transfer functions and subsequent equalisation with appropriate digital filters. As a result, the loudspeaker did not differ by more than 1dB below 400Hz and 0.5dB above. Informal listening test with broad-band noise did not show audible differences between the equalised loudspeakers.

For the use in virtual-reality systems, the HRTFs have to be accompanied by the transfer functions of the headphones on the ear, measured at the same microphone positions where the HRTFs have been measured. Here, headphone transfer functions have been determined for STAX-Lambda-Pro headphones for each individual ear under test.

Main differences of the Soesterberg set-up as compared to Bochum are as follows: Instead of a rotation chair, a loudspeaker-positioning system was used. The subject sat in a fixed position, monitored by an inductive head tracker. The loudspeaker, mounted on a carriage running on a $\frac{3}{4}$ rotating circle, could be positioned to sample an (almost) complete sphere. In fact, 965 positions were actually used for the measurements reported here. The distance of the loudspeaker from the head centre was 1.16m. The measuring signal was a 10.2ms chirp, repeated 25 times. The room's lower limiting frequency was 200Hz. The microphones were placed directly in front of the eardrums.

ROUND-ROBIN TEST

To check on the accuracy and reliability of HRTF measurement, a round-robin test has been executed, incorporating two further laboratories, namely, Head acoustics at Herzogenrath and CERMA at Paris - the latter being commissioned by SEXTANT. In the tests, identical subjects as well as two artificial heads were measured with different procedures. The outcome of these tests resulted in some "Golden Rules" for HRTF measurements, which will be published elsewhere (e.g., on the sampler CD mentioned). The measurements performed at Bochum and Soesterberg comply with these rules to a high degree. For highest requirements in terms of physical accuracy, the head position should be monitored by a head tracker, and the measuring chain should be corrected on line, accordingly.

PERCEPTUAL VERIFICATION

It goes without saying that the ultimate evaluation of the quality HRTFs has to be performed auditorily, not physically. To this end, perceptual tests have been executed with virtual sound sources, synthesised with the individual HRTFs and headphone transfer functions measured. In other words, the subjects were presented with artificial signals which represent our best efforts to replicate the input signals to the two ears in the real situation to be simulated for static conditions - which means that head movements were not catered for. In comparison, the signals of the "real" sources were also presented - again with the head in a fixed position - and the subjects had to do a localisation task for both the real and the corresponding virtual sound sources.

In each case, it was the task of the subject to determine the direction of the auditory events. For indication of the perceived directions, an inductive position tracker was used as a pointer on an acrylic sphere of 15cm diameter (We use to call this device the Bochum Sphere, as it was developed well before its commercial counterpart). Tests were performed for positions in different directions of the upper hemisphere. In summary, the outcome is as follows: Simulated (virtual) sound sources are localised very much like real ones, when individual HRTFs and headphone transfer functions are being used. For the horizontal plane, deviations between the directional perception of real and virtual sources are minimal. As to elevation, there are slightly higher deviations at higher elevations. Front back reversals are, on an average, more than twice as frequent as with real sources. Mind, however, that a fixed-head condition was used in all the experiments reported here, that is, there was no account of head movements.

VARIA

To make efficient use of HRTFs, the user may wish to obtain some signal-processing tools with the data. To this end the following items have been developed and will be available together with the HRTF catalog: A tutorial-like introduction which presents different auralisations of multi-talker scenarios, from simple stereo panning through HRTF-based synthesis to full binaural room simulation. Further, there will be tools for graphic representations of HRTFs, both in the time and frequency domain. The HRTFs cannot only be viewed, but also be convolved with "dry" mono signals - some of which will be provided as well. For example, the sampler CD will contain some dry chamber music (flute, violin cello, harpsichord) and dry speech of different speakers. As the CD cannot provide data for all possible directions of sound incidence, there will be tools to interpolate between different measured directions, in order to enable the user to try to make an approximation of HRTFs for directions not provided in the catalog. To achieve an extended auditory spatial impression, the CD will also include some binaural room-impulse responses. In detail, there will be a choice of binaural impulse responses measured in real, typical concert halls (e.g., a shoe-box-type one and a V-shaped one) as well as some typical binaural impulse responses as rendered by the Computer-Aided Room-Simulation system, CARS, of the Institute of Communication Acoustics at Bochum.